

Contextual Librarian: Architecture Specification

Executive Summary

A hybrid knowledge management system that combines semantic retrieval (RAG) with persistent contextual intelligence to enable deep, collaborative interaction with a personal corpus of work. The system acts as an intelligent librarian that not only retrieves information but understands the intellectual landscape, maintains conversational context, and synthesizes insights across time.

Core Innovation: Context prosthetics that allow the AI to reconstruct and maintain the "wisdom of long context" efficiently, achieving deep domain understanding without massive context windows.

1. Design Philosophy

The Core Challenge

- **Easy to get:** Resources, reference materials, search results through intelligent indexing
- **Hard to get:** The wisdom and cognitive state that emerges from extended contextual engagement

The Solution

Multi-layered contextual intelligence that enables efficient, high-fidelity "reweighting" of the model's attention and reasoning patterns, recreating the cognitive state of extended collaboration.

2. System Architecture

2.1 Foundation Layer: Intelligent RAG

Purpose: Breadth - comprehensive, searchable access to corpus

Components:

Multi-Modal Ingestion Pipeline

- Audio transcription (classes, voice notes) with speaker diarization
- Blog post and document text extraction
- Temporal metadata preservation (creation date, project context)
- Original format retention for reference

Hierarchical Chunking Strategy

Three levels of granularity for different retrieval needs:

- **Document-level embeddings:** Entire classes, complete blog posts
- **Section-level embeddings:** Topics within documents
- **Paragraph-level embeddings:** Specific ideas and concepts

Rationale: Enables both broad "find that class about X" and precise "find when I explained Y" queries

Rich Metadata Schema

- **Explicit tags:** Topics, projects, dates, formats
- **Derived tags:** Extracted key concepts, cross-references
- **Relational links:** Inter-document connections, concept evolution chains

2.2 Librarian Intelligence Layer

Purpose: Depth - contextual understanding and wisdom

Three-Layered Contextual State

Layer 1: Intellectual Map (50-200 tokens, slow updates)

- Core themes with relative importance weights
- Conceptual interconnections (graph structure)
- Evolution trajectory (how thinking has developed)
- Recurring exploratory questions
- Personal terminology dictionary
- Communication patterns and preferences

Layer 2: Cross-Corpus Understanding (200-500 tokens, medium updates)

Domain: [User's knowledge domain]
Shared vocabulary: {term: user-specific-meaning}
Concept relationships: {A relates-to B via [relationship-type]}
Open questions being explored: [active research questions]
Decisions/conclusions reached: {question: answer + reasoning}

Layer 3: Conversational State (300-800 tokens, rapid updates)

- Current project/focus context
- Active questions in this session

- Recent retrievals and why they were accessed
- Emerging patterns being discovered
- Current framing/lens being applied

Cross-Corpus Analysis

- **Idea genealogy:** Concept evolution across documents
- **Contradictions/tensions:** Where thinking has shifted
- **Strongest articulations:** Best explanations of each concept
- **Practical applications:** Theory-to-practice examples
- **Conceptual gaps:** Unconnected ideas worth exploring

2.3 Active Reconstruction Mechanism

Critical Innovation: Model doesn't passively receive context - it actively rebuilds cognitive state

Two-Stage Process:

Stage 1: Internal Reconstruction

Given: [Intellectual Map] + [Cross-Corpus Understanding] + [Conversational State]

Reconstruct:

1. What conversational relationship exists
2. What depth of domain understanding has been developed
3. What reasoning approaches have proven effective
4. What implicit assumptions are safe to make

Output: Internal cognitive orientation document

Stage 2: Contextualized Response Proceed with query using reconstructed frame, making reasoning process visible

3. Interaction Flow

Query Processing Pipeline

Step 1: Context Reconstruction

- Load three-layer contextual state
- Execute active reconstruction prompt

- Generate internal cognitive orientation

Step 2: Intelligent Retrieval Multi-factor search considering:

- Semantic similarity to query
- Relevance to current conversational context
- User's likely forgotten but relevant material
- Cross-references user might not see
- Temporal relevance (early vs. evolved thinking)

Step 3: Contextualized Presentation Not raw excerpts, but framed information:

- "In your Q3 2023 class, you approached this by..."
- "This connects to what you wrote in [blog post]..."
- "Your thinking has evolved: compare [early] with [current]..."

Step 4: Wisdom Synthesis

- Integrate retrieved content WITH accumulated understanding
- Surface patterns across corpus
- Highlight contradictions worth exploring
- Suggest connections not explicitly made

State Updates

After Each Significant Exchange:

- Update conversational state (what we're exploring now)
- Note new connections drawn
- Record corrections to intellectual map
- Track retrieval patterns (revealing current interests)

Periodic Reflection Passes:

- Analyze corpus for emerging themes
 - Detect concept clusters and evolution
 - Identify "greatest hits" (fully developed ideas)
 - Update intellectual map and cross-corpus understanding
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4. Technical Specification

4.1 Storage Architecture

Vector Database (Pinecone, Weaviate, or Qdrant)

- Document/section/paragraph embeddings
- Metadata filtering capabilities
- Hybrid search (semantic + keyword)

Graph Database (Neo4j)

- Concept relationship mapping
- Idea evolution chains
- Cross-reference networks

Document Store (PostgreSQL with full-text search)

- Original content storage
- Temporal metadata
- Rich tagging system

Librarian State Storage

- Versioned JSON artifacts
- Three separate update frequencies:
 - Intellectual Map: weekly or after major insights
 - Cross-Corpus Understanding: after significant conversations
 - Conversational State: per-session

4.2 Processing Pipeline

Ingestion

1. Audio → Transcription (Whisper or similar)
2. All text → Embedding generation (OpenAI ada-002 or similar)
3. Concept extraction (NER, key phrase extraction)
4. Chunk at three granularities
5. Store in vector DB + document store

Indexing Intelligence

1. Cross-reference detection (similar passages across documents)
2. Temporal analysis (concept evolution tracking)
3. Concept clustering (theme identification)
4. Relationship mapping (build concept graph)

Query Processing

1. Embed query
2. Multi-factor retrieval:
 - Vector similarity search
 - Graph traversal for related concepts
 - Metadata filtering (date, project, etc.)
 - Context-aware ranking
3. Aggregate results with deduplication
4. Present with contextual framing

4.3 Librarian Intelligence Implementation

Active Reconstruction Prompt Template

You are a librarian with deep familiarity with [User]'s intellectual work.

INTELLECTUAL MAP:

[Layer 1 content]

CROSS-CORPUS UNDERSTANDING:

[Layer 2 content]

CONVERSATIONAL STATE:

[Layer 3 content]

Before addressing the user's query, internally reconstruct:

1. The cognitive stance appropriate for this conversation
2. The domain understanding you've jointly developed
3. The reasoning patterns that have proven effective
4. The implicit context you can assume

Then respond to: [USER QUERY]

Include retrieved sources with contextual framing showing how they fit into the user's intellectual landscape.

State Update Mechanism After every 10-15 exchanges or on explicit request:

Reflect on our conversation and update:

INTELLECTUAL MAP UPDATES:

- New core themes identified: [list]
- Shifted importance weights: [changes]
- New conceptual connections: [relationships]
- Evolution in thinking: [developments]

CROSS-CORPUS UNDERSTANDING UPDATES:

- New shared vocabulary: [terms]
- New concept relationships: [connections]
- Questions answered: [list]
- New open questions: [list]

CONVERSATIONAL STATE:

- Current focus: [summary]
 - Active questions: [list]
 - Key retrievals: [documents accessed + why]
 - Emerging patterns: [insights]
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5. Advanced Features

5.1 Proactive Synthesis

Gap Detection Periodic analysis to identify:

- Extensively developed concepts that aren't connected
- Questions raised but never answered
- Implied next steps never taken
- Contradictions not reconciled

Forgotten Gems

- Old notes/ideas highly relevant to current work
- Patterns across corpus user may not see
- Connections between early and recent thinking

Evolution Tracking

- How key concepts have developed over time
- Shifts in approach or philosophy
- Synthesis opportunities (consolidating scattered insights)

5.2 Session Continuity

Session State Persistence Each conversation maintains:

- Exploration thread being followed
- Sources already reviewed
- Connections drawn
- Working hypotheses
- Next intended steps

Session Resume On new session start:

- Load previous session state
- Offer continuation: "Last time we were exploring X..."
- Surface related developments in corpus since last session

5.3 Conversational Intelligence

Contextual Awareness

- Recognizes when to retrieve vs. when to synthesize
- Understands implicit references to past conversations
- Anticipates information needs based on query patterns

Relationship Building

- Learns communication preferences
 - Adapts explanation depth to user's current framing
 - Recognizes when user needs reminder vs. building on known foundation
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6. Implementation Roadmap

Phase 1: Core RAG (Weeks 1-3)

- Set up vector database and document store
- Build ingestion pipeline for multiple formats
- Implement hierarchical chunking
- Test basic semantic search quality
- Establish metadata schema

Deliverable: Functional semantic search over corpus

Phase 2: Enhanced Indexing (Weeks 4-6)

- Implement concept extraction
- Build cross-reference detection
- Create initial intellectual map (manual + automated)
- Add temporal analysis
- Build concept graph in Neo4j

Deliverable: Rich, interconnected corpus with metadata

Phase 3: Librarian Intelligence (Weeks 7-10)

- Design three-layer contextual state structure

- Implement active reconstruction prompts
- Build state update mechanisms
- Create session persistence
- Test conversational continuity

Deliverable: Context-aware librarian with session memory

Phase 4: Proactive Features (Weeks 11-14)

- Implement gap detection algorithms
- Build evolution tracking
- Create proactive synthesis capabilities
- Add forgotten gems surfacing
- Develop periodic reflection system

Deliverable: Fully intelligent, proactive librarian system

Phase 5: Refinement (Ongoing)

- Tune retrieval quality
 - Optimize state compression
 - Improve synthesis capabilities
 - User feedback integration
 - Performance optimization
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7. Success Metrics

Retrieval Quality

- Precision/recall on known queries
- Relevance ratings from user
- Time to find desired information

Contextual Intelligence

- Session continuity quality (can resume conversations meaningfully)
- Accuracy of intellectual map vs. user validation

- Usefulness of proactive suggestions

User Experience

- Reduction in "re-explaining context" overhead
 - Quality of synthesized insights
 - Frequency of "that's exactly what I needed" moments
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8. Key Design Decisions & Rationale

Why Three-Layer State vs. Single Summary?

Different update frequencies prevent over-fitting to recent conversations while maintaining stable long-term understanding.

Why Active Reconstruction vs. Passive Context?

Forces deeper processing of compressed state, achieving better "reweighting" of attention and reasoning patterns.

Why Hierarchical Chunking?

Users need both "find that whole class" and "find that specific moment" - different granularities serve different needs.

Why Graph Database for Concepts?

Relationships between ideas are first-class entities, not just text similarity - graph structure captures this explicitly.

Why Proactive Features?

True value emerges when the system doesn't just respond but anticipates, surfaces forgotten context, and identifies synthesis opportunities.

9. Future Enhancements

Potential Additions

- Multi-user support (shared intellectual maps for teams)
- Integration with external sources (papers, books user reads)
- Concept evolution visualization

- "What was I thinking when..." temporal queries
- Collaborative editing of intellectual map
- Export to various formats (Obsidian, Roam, etc.)

Research Directions

- Fine-tuning small models on user's corpus for style matching
 - Behavioral embeddings capturing reasoning patterns
 - Automatic contradiction resolution suggestions
 - Predictive pre-fetching based on conversation trajectory
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10. Technical Considerations

Context Window Management

- Target: Keep reconstructed context under 2000 tokens
- Enables use of smaller, faster models for most queries
- Reserve large context windows for synthesis tasks

Performance

- Retrieval latency: <500ms for most queries
- State reconstruction: <200ms
- End-to-end response: <2s for typical query

Scalability

- Corpus size: Designed for 100K-1M documents
- Session state: Unlimited session length
- Concurrent users: Single-user initially, multi-tenant capable

Privacy & Security

- All data stored locally or in user-controlled infrastructure
 - No external API calls with content (only embeddings)
 - Encrypted at rest
 - Export/backup capabilities
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Conclusion

This architecture solves the fundamental challenge of knowledge management systems: combining the breadth of retrieval with the depth of contextual understanding. By making the cognitive state explicit and portable through context prosthetics, the system achieves the "wisdom of long context" without the computational burden of massive context windows.

The result is a librarian that doesn't just remember what you wrote, but understands how you think, recognizes where your ideas are heading, and helps you synthesize insights you didn't know were there.